Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore

Shri Vaishnav Institute of Information Technology

Choice Based Credit System (CBCS) in the light of NEP-2020

B.Tech./ B. Tech + M.Tech / B.Tech + MBA (CSE/ IT/CSBS) - All Programs

SEMESTER-VII (2023-2027)

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			THEORY			PRACTICAL					
			END SEM University Exam	Two Term Exam	Teachers Assessment*	END SEM University Exam	Teachers Assessment*				
BTCS706N	SEC	Project	0	0	0	120	80	0	0	8	4

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COURSE OBJECTIVES:

The student will have ability to:

1. Enable students to identify real-world, complex engineering problems and apply systematic research methodologies to formulate viable technical solutions.
2. Train students in designing, engineering, testing, and validating a complete product, system module, or computing architecture using modern engineering tools and industry-standard practices.
3. Foster collaborative engineering management skills, technical report writing, and the professional defense of engineered solutions before an evaluation panel.

COURSE OUTCOMES:

Upon completion of the subject, students will be able to:

1. Formulate a well-defined problem statement by critically reviewing contemporary research literature, identifying technical gaps, and establishing clear engineering requirements.
2. Design and develop an architectural framework, hardware prototype, or software system by incorporating sustainable design methodologies, algorithm design, and resource optimization.
3. Implement and deploy the proposed system using modern development environments, programming paradigms, cloud infrastructure, or advanced technical tools.
4. Evaluate the performance, scalability, and security parameters of the developed system against standard benchmark metrics, interpreting the results through rigorous testing.
5. Compile comprehensive, industry-grade technical project reports and documentation compliant with standard engineering formats, defending the work effectively through oral presentations.

Project Timeline

Phases

Phase 1: Planning & Requirement Analysis (Weeks 1–2)

Focus: Defining the scope, setting up infrastructure, and identifying functional requirements.

- **Week 1: Project Initiation**
  - Form teams and define the problem statement.
  - **Tooling:** Set up **Jira** boards (Scrum or Kanban), create Backlog, and initialize **Git** repositories.
  - **Deliverable:** Project Vision document and initial Product Backlog.

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- **Week 2: Functional Modeling**
  - Identify Actors and Use Cases.
  - **UML Deliverables: * Use Case Diagram:** To define system boundaries.
    - **Use Case Descriptions:** Detailed text for each primary flow.
  - **Tooling:** Map Use Cases to Jira User Stories.

Phase 2: Object-Oriented Analysis (Weeks 3–4)

Focus: Understanding the "What" of the system without focusing on implementation.

- **Week 3: Static Analysis**
  - Identify domain entities and their relationships.
  - **UML Deliverables:**
    - **Object Diagram:** To model specific instances and data snapshots.
    - **Initial Class Diagram (Domain Model):** Focus on attributes and relationships (1:1, 1:N, M:N) without methods.

- **Week 4: Dynamic Analysis**
  - Model how the system reacts to events.
  - **UML Deliverable: Activity Diagram:** To model complex business logic and parallel workflows.

Phase 3: Object-Oriented Design (Weeks 5–7)

Focus: Defining the "How" – architecture, patterns, and component interactions.

- **Week 5: Interaction Modeling**
  - **UML Deliverable: * Sequence Diagram:** Detailed logic showing how objects communicate over time to fulfill a use case.

- **Week 6: Detailed Class Design & Packaging**
  - Apply SOLID principles and Design Patterns.
  - **UML Deliverables:**
    - **Detailed Class Diagram:** Including methods, visibility, and interfaces.
    - **Package Diagram:** Organizing classes into logical modules/subsystems.

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- **Week 7: System Architecture**
  - **UML Deliverables:**
    - **Component Diagram:** Visualizing physical modules (API, Database, UI).
    - **Deployment Diagram:** Mapping software components to hardware/nodes (Cloud, Servers).

Phase 4: Implementation & CI/CD Integration (Weeks 8–10)

Focus: Coding, version control, and automated pipelines.

- **Week 8: Sprint 1 - Core Features**
  - Develop core logic based on the Class and Sequence diagrams.
  - **Git Strategy:** Use branching (Feature branches → Develop → Main).

- **Week 9: CI/CD Pipeline Setup**
  - Configure **GitHub Actions** or **GitLab CI**.
  - Automate build triggers on every git push.

- **Week 10: Sprint 2 - Integration**
  - Develop UI and connect to the database.
  - **Jira:** Move stories to "In Progress" and "Testing."

Phase 5: Testing & Deployment (Weeks 11–12)

Focus: Quality assurance and final delivery.

- **Week 11: Testing Phase**
  - **Unit Testing:** (JUnit, PyTest) ensuring methods work as intended.
  - **Integration Testing:** Testing the flow between components.
  - **Tooling:** Integrate test reports into the CI/CD pipeline.

- **Week 12: Deployment & Final Review**
  - Finalize the **Deployment Diagram** to match the production environment.
  - Deploy to a staging/production environment (Heroku, AWS, or Azure).
  - Project Demo and handover.

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Jira Backlog and a **Git Branching Strategy**

1. Jira Backlog & Sprint Structure

The project is divided into **six 2-week Sprints**. **Sample Jira Epic & Story Structure:**

Epic	Sample User Stories (Tasks)	Status
Requirements & Design	Create Use Case Diagrams; Draft System Architecture; Set up Jira Board.	Done/Sprint 1
Infrastructure	Initialize Git Repo; Configure CI/CD Pipeline; Setup Database.	Sprint 2
User Management	Design User Class; Implement Login/Auth; Sequence Diagram for Auth.	Sprint 3
Core Features	[Project Specific Logic]; Unit Tests for Core Logic.	Sprint 4-5
Deployment	Final Deployment Diagram; Production Hosting; Documentation.	Sprint 6

2. Git Branching Strategy

Students should avoid pushing directly to the main branch. A **Feature Branch Workflow** is best for academic teams to prevent code conflicts.

- **main branch:** Contains stable, production-ready code.
- **develop branch:** The integration branch where all features are merged before going to main.
- **feature/feature-name branches:** Created for every individual Jira task (e.g., feature/user-login).

3. Testing & CI/CD Pipeline

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To meet your requirement for testing software and CI/CD, the students should implement a pipeline that triggers on every **Pull Request** to the develop branch.

The Pipeline Flow:

1. **Build:** Compile the code to ensure there are no syntax errors.
2. **Unit Test:** Automatically run tests (e.g., JUnit, PyTest, or Jest). If tests fail, the merge is blocked.
3. **Static Analysis:** Use tools like SonarQube or ESLint to check for code "smells."
4. **Deploy:** Automatically deploy to a staging environment (like Vercel, Netlify, or AWS).

4. UML to Code Mapping (Object-Oriented Focus)

Students must ensure that their diagrams are not just "drawings" but blueprints:

- **Class Diagram** → Directly maps to the Classes and Attributes in their code.
- **Sequence Diagram** → Maps to the logic inside their methods (which object calls which method).
- **Package Diagram** → Maps to the Folder Structure of their project.

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Project Grading Rubric (Total 100%)

1. Analysis & Design (30%)

- **UML Consistency:** Do the diagrams match the actual code implementation?
    - *Check:* Does the **Class Diagram** reflect the actual inheritance and associations in the source code?
- **Behavioral Accuracy:** Do **Sequence Diagrams** correctly identify method calls between objects?
- **Architectural Soundness:** Does the **Deployment Diagram** accurately map to the cloud/server infrastructure used?

2. Object-Oriented Implementation (25%)

- **Encapsulation:** Are attributes private? Are getters/setters used appropriately?
- **Inheritance/Polymorphism:** Are these used to reduce code redundancy?
- **Solid Principles:** Is the code modular? (e.g., Single Responsibility Principle).
- **Design Patterns:** Did the team implement at least one recognizable pattern (e.g., Singleton, Factory, or Observer)?

3. DevOps & Tooling (20%)

- **Jira Management:** Is the backlog updated? Are stories assigned to specific members?
    - *Check:* Do commit messages reference Jira ticket IDs?
- **Git Workflow:** Is there evidence of Pull Requests and Code Reviews?
    - *Check:* Avoid "Mega-commits" (one giant commit at the end of the week).
- **CI/CD Pipeline:** Does the pipeline automatically trigger on pushes? Is the build status visible (Green/Red)?

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4. Testing & Quality (15%)

- **Unit Testing:** Is there a significant percentage of code coverage?
- **Integration Testing:** Are API endpoints or database connections tested?
- **Documentation:** Is the README.md clear, including setup instructions and a system overview?

5. Final Deployment (10%)

- **Accessibility:** Is the project hosted on a live URL?
- **Stability:** Does the system handle basic "unhappy path" errors without crashing?

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Definition of Done (DoD) Checklist

A Jira ticket is only considered **Done** when it meets the following criteria:

1. Design & Modeling

- **UML Alignment:** Any changes to the code structure are reflected in the **Class Diagram**.
- **Logic Validation:** For complex logic, a **Sequence Diagram** has been updated or created.
- **Documentation:** Public methods and classes are documented using standard docstrings (e.g., Javadoc, Docstring).

2. Code Quality & Standards

- **OO Principles:** Code follows Encapsulation and avoids "God Classes" (Single Responsibility Principle).
- **No Hardcoding:** Configuration values (like API keys or DB URLs) are in environment variables.
- **Peer Review:** At least one other team member has reviewed the **Pull Request (PR)** on Git.

3. Testing & CI/CD

- **Unit Tests:** New logic has corresponding unit tests.
- **CI Pipeline:** The CI/CD build is **Green** (passing) on the feature branch.
- **Zero Regressions:** Existing functionality still works as expected.

4. Integration

- **Git Merge:** The feature branch has been merged into the develop branch.
- **Jira Update:** The ticket is moved to the "Done" column and the "Actual Time Spent" is logged.

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Evaluation Schedule

Phase	Milestone	Expected Artifacts
Month 1	**Foundation**	Use Case, Activity, and Object Diagrams. Jira Backlog setup.
Month 2	**Design**	Sequence, Class, and Package Diagrams. Git Feature branches active.
Month 3	**Execution**	Component and Deployment Diagrams. Working CI/CD pipeline and Unit Tests.

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Weekly Progress Report Template

Project Title: [Project Name]

Team ID: [ID]

Date: [Date Range]

1. Executive Summary

- **Current Sprint:** [Sprint # of 6]
- **Sprint Goal:** (e.g., Finalize Sequence Diagrams and setup Database connection)
- **Status:** On Track | At Risk | Delayed

2. UML & OOAD Progress

- **Diagrams Completed this week:** (e.g., Package Diagram, Component Diagram)
- **Diagrams Revised:** (List any changes made based on implementation feedback)
- **OO Concept Applied:** (e.g., "Implemented Interface Inheritance for the Payment module")

3. Technical Execution (DevOps & Git)

- **Jira Tasks Completed:** [Link to Jira tickets]
- **Git Activity:**
  - Active Branches: [List branch names]
  - Merged Pull Requests: [List PR IDs]
- **CI/CD Status:** [e.g., Build Passing / 85% Code Coverage]

4. Testing Summary

- **Total Unit Tests:** [Number]

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- **Pass Rate:** [Percentage]%
- **Key Bug Identified:** [Brief description of one technical challenge overcome]

5. Goals for Next Week

- [Goal A]
- [Goal B]

Project Guide Signature

Project In-charge Signature